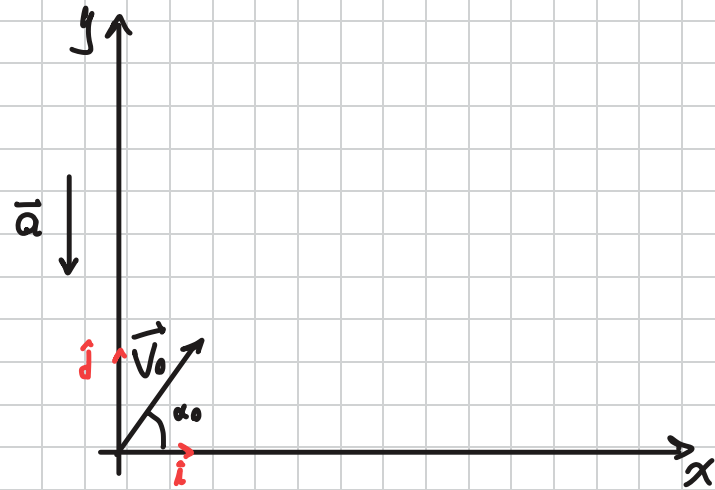




Projectile motion



$$a_x = 0, a_y = -g$$

initial position $x_0 = 0, y_0 = 0$

initial velocity
$$\vec{V}_0 = V_0 \cos \alpha_0 \hat{i} + V_0 \sin \alpha_0 \hat{j}$$

X: $a_x = 0 \Rightarrow$ Constant velocity

$$x = x_0 + v_{0x} t = v_0 \cos \alpha_0 t \quad (1)$$

Y: Constant acceleration $a_y = -g$

$$v_y = v_{0y} + a_y t = v_0 \sin \alpha_0 - gt \quad (2)$$

$$y = y_0 + v_{0y} t + \frac{1}{2} a_y t^2 = v_0 \sin \alpha_0 t - \frac{1}{2} g t^2 \quad (3)$$

Solve for t from eqn (1)

$$t = \frac{x}{v_0 \cos \alpha_0} \quad \text{Sub into (3)}$$

$$y = \cancel{V_0} \sin \alpha_0 \frac{x}{\cancel{V_0} \cos \alpha_0} - \frac{1}{2} g \frac{x^2}{V_0^2 \cos^2 \alpha_0}$$

$$y = \tan \alpha_0 x - \frac{g}{2V_0^2 \cos^2 \alpha_0} x^2$$

Equation of the form $y = ax - bx^2$
has a graph in the shape of a parabola

Require that $y=0$

$$\tan \alpha_0 x - \frac{g}{2V_0^2 \cos^2 \alpha_0} x^2 = 0$$

$$x \left(\tan \alpha_0 - \frac{g}{2V_0^2 \cos^2 \alpha_0} x \right) = 0$$

One solution is $x=0$ (launch point)

the other: $\tan \alpha = \frac{g}{2V_0^2 \cos^2 \alpha} X$

$$X = \frac{2V_0^2 \cos^2 \alpha_0 \tan \alpha_0}{g} = \frac{2V_0^2 \sin \alpha_0 \cos \alpha_0}{g}$$

or $x_{\max} = \frac{V_0^2}{g} \sin 2\alpha_0$

For maximum distance traveled (horizontally)

we must have $\sin 2\alpha_0 = 1$

$$\Rightarrow 2\alpha_0 = \frac{\pi}{2} \Rightarrow \alpha_0 = \frac{\pi}{4}$$

optimize launch
angle

What is the
maximum height?

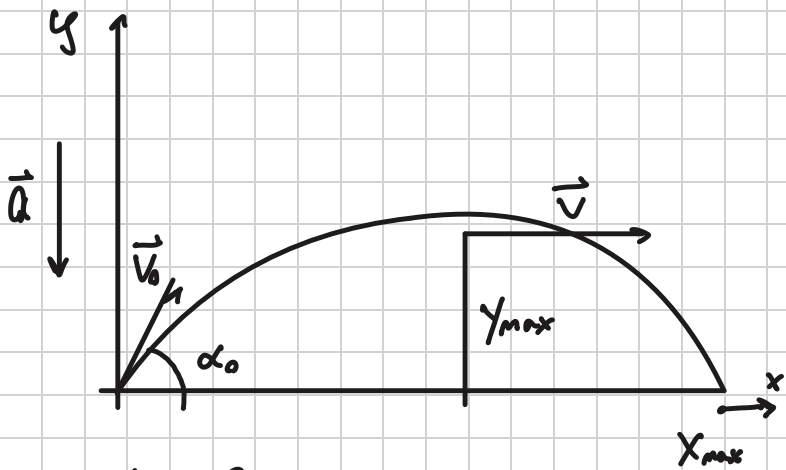
At maximum height, $V_y = 0$

$$V_y^2 = V_{0y}^2 + 2ay \Rightarrow V_y = 0, a = -g$$

$$V_{0y} = V_0 \sin \alpha_0$$

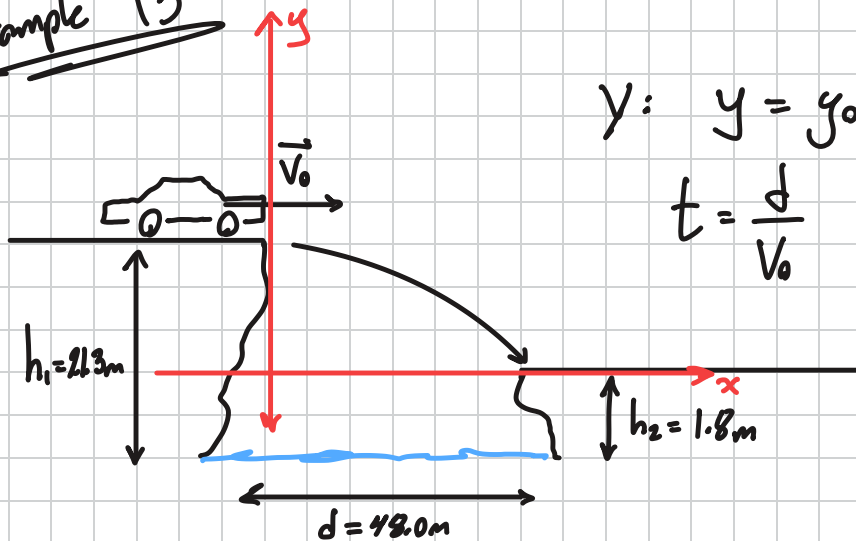
$$0 = V_0^2 \sin^2 \alpha_0 - 2g y_{\max} \Rightarrow$$

$$y_{\max} = \frac{V_0^2 \sin^2 \alpha_0}{2g}$$



$$y_0 = h_1 - h_2$$

Example 13



$$x: x = v_0 t$$

$$y: y = y_0 + v_{0y} t + \frac{1}{2} a t^2 = (h_1 - h_2) - \frac{1}{2} g t^2$$

$$t = \frac{d}{v_0}$$

time for y to drop to zero:

$$(h_1 - h_2) - \frac{1}{2} g t^2 = 0$$

$$t^2 = \frac{2(h_1 - h_2)}{g}$$

Set the two equal

$$t = \sqrt{\frac{2(h_1 - h_2)}{g}}$$

$$\frac{d}{V_0} = \sqrt{\frac{2(h_1 - h_2)}{g}} \Rightarrow V_0 = d \sqrt{\frac{g}{2(h_1 - h_2)}}$$